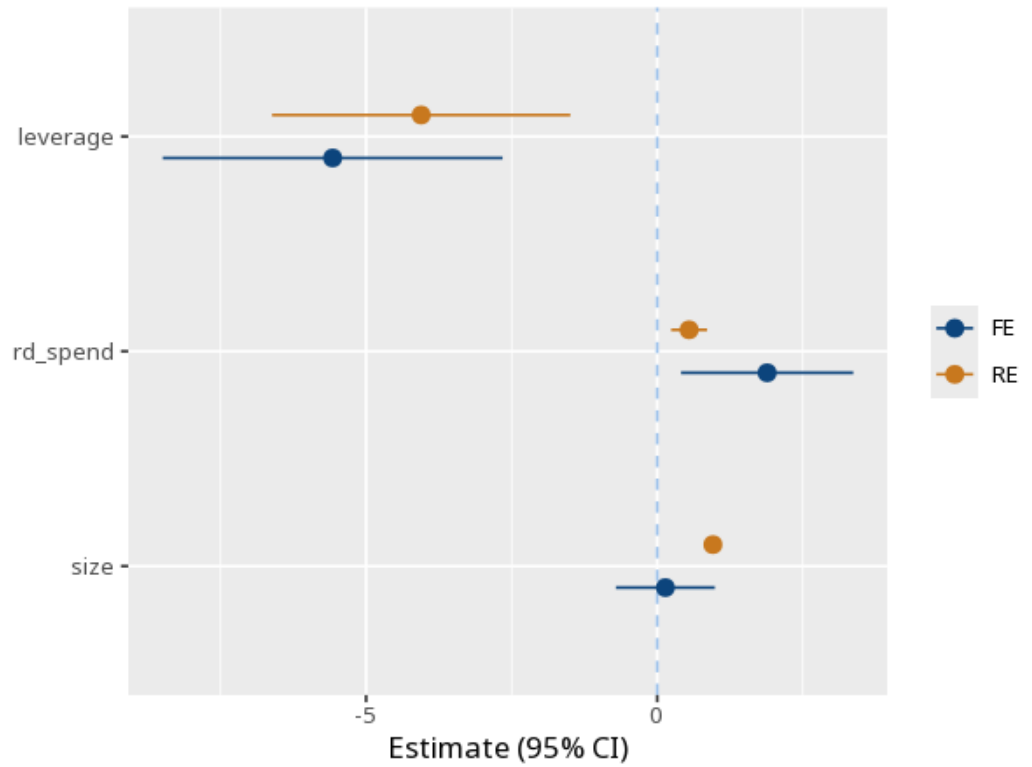


1 Hausman test

	Fixed effects	Random effects	Difference
leverage	−5.57 (1.47) [−8.49, −2.66]	−4.05 (1.29) [−6.62, −1.49]	−1.52
rd_spend	1.89 (0.74) [0.41, 3.37]	0.55 (0.16) [0.24, 0.86]	1.34
size	0.14 (0.43) [−0.71, 0.99]	0.96 (0.05) [0.86, 1.05]	−0.82
Num.Obs.	96		
N entities	12		
R ²	.91	.98	
Hausman $\chi^2(3) = 3.07$, p .381			



Tests whether the random-effects assumption holds. A p below alpha means the estimates differ systematically → use fixed effects; a non-significant p means random effects is acceptable (and more efficient). Method: R plm package (Croissant & Millo, 2008). The standard errors in parentheses are clustered by

entity; the bracketed line is the 95% CI. Your significance threshold (α) is 0.05.

A Hausman test comparing the fixed- and random-effects estimates gave $\chi^2(3)=3.07$, $p = .381$.

Statistical basis: Hausman specification test (Hausman, J. A. (1978). "Specification tests in econometrics." *Econometrica*, 46(6), 1251-1271.); Implementation (plm package) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." *Journal of Statistical Software*, 27(2), 1-43.).